

0 8

Alport syndrome (AS) is an inherited disorder that affects kidney glomeruli of both men and women. Affected individuals have proteinuria (high quantities of protein in their urine).

0 8**. 1**

Suggest how AS could cause proteinuria.

[2 marks]

0 8**. 2**

AS results from a sex-linked mutation.

In a male with AS, where would the sex-linked mutation be located?

Tick (✓) **one** box.

[1 mark]

The homologous section of a Y chromosome

☐

The homologous section of an X chromosome

☐

The non-homologous section of a Y chromosome

☐

The non-homologous section of an X chromosome

☐

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The scientists carried out further work to investigate how the transplanted stem cells developed after transplantation.

- The scientists transplanted stem cells from wild type male mice into AS female mice.
- After 20 weeks, they found that the quantity of protein in the urine of these female mice had significantly decreased.
- They examined cells from glomeruli in the female mice. Some of these cells contained a Y chromosome.

Suggest how the transplanted stem cells reduce proteinuria.

[2 marks]

Turn over for the next question

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write outside*

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ANSWER IN THE SPACES PROVIDED**

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